



INDIVIDUAL ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT018-3-1-ICP

INTRODUCTION TO C PROGRAMMING

APU1F2209CS

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1. Introduction and Assumptions

APU have made a decision to launch a program of experimentation named "Programming Café". This program is to offer extra coding sessions for students outside of their usual timetable in order to boost their performance, increase their problem-solving abilities, and boost their confidence level in coding. There will be a classroom provided for each session, and each session are two hours long. Programming Café sessions will be guided by selected APU Graduate Students who are proficient in their respective programming languages. They will get a set charge of RM100 for each Two-hour session. In this stage of testing, APU has chosen a few popular programming languages for the Programming Café Sessions. Table 1 contains a list of the chosen programming language sessions. If more time is needed, more sessions on other coding topics can be added. But every session shown in Table 1 has to be added to the system by default.

Table 1: Programming Café Sessions

Session Code	Title	Day	Start Time	Location	Tutor Code
PYP101	Python Programming	Saturday	9.00am	C-01-01	T01
JAV102	Java Programming	Sunday	9.00am	C-01-02	T02
CPL103	C Programming	Saturday	2.00pm	C-01-03	T03
WEB104	Web Development	Sunday	2.00pm	C-01-04	T04
CSP105	C Sharp Programming	Monday	7.00pm	C-01-05	T05

Table 2: Tutor

Role	User Code	Name
Tutor	T123456	Danny
Tutor	T789123	YitWah
Tutor	T456789	Joanne
Tutor	T101112	Linda

2. Design of the program (Pseudocode)

Record structures

```

sessions
    session_code
    title
    day
    start_time
    location
    tutor_code

enroll_sessions
    user_code
    user_name
    user_role
    session_code

users
    session_title
    user_role
    user_code
    user_ame
    email
    password
  
```

Module declarations

```

/* int main_menu () */
main_menu
    SET choice AS integer

    PRINT ("-----")
    PRINT ("|   APU Programming Cafe Management System   |")
    PRINT ("|-----|")
    PRINT ("|               >>  HOME PAGE  <<               |")
    PRINT ("|-----|")

    PRINT ("1. Login")
    PRINT ("2. View Available Programming Session")
    PRINT ("3. Exit the Programming Cafe Session")
    PRINT ("Enter your choice:")

    READ choice
    RETURN choice
END
  
```

```
/* int main() */
main
    create_file

    SET flag = 1 (TRUE)
    SET choice AS integer
    SET authenticated AS integer

    DOWHILE flag = 1 (TRUE)
        SET choice = main_menu
        CASE OF CHOICE
            1:
                SET authenticated = login_page
                IF authenticated = 0 THEN
                    SET flag = 0
                ENDIF
                BREAK
            2:
                Session_menu
                BREAK
            3:
                SET flag = 0
                PRINT("~~~ Thanks for using APU Programming Cafe Management System ~~")
                BREAK
            DEFAULT:
                PRINT (" !!! Invalid option, Please try again !!! ")
                PRINT (" ----- ")
        ENDCASE
    ENDDO
    RETURN 0
END
```

```

/* int login_page() */
login_page
    PRINT ("-----")
    PRINT ("|    APU Programming Cafe Management System    |")
    PRINT ("|-----|")
    PRINT ("|                >> LOGIN PAGE <<                |")
    PRINT ("|-----|")

    SET users_file = OPEN FILE (user_profile.txt) WITH "r" mode

    SET authenticated = 0
    SET login_attempts = 0
    SET max_login_attempts = 3

    DOWHILE login_attempts < max_login_attempts
        REWIND THE FILE POINTER TO BY users_file TO THE BIGINING
        SET structure users AS user
        SET user_code WITH MAXIMUM LENGTH Max_User_Code
        SET password WITH MAXIMUM LENGTH Max_String

        PRINT("Please Enter Your User Code:")
        GET INPUT user_code
        PRINT("Password: ")
        GET INPUT password

        SET response AS integer, user_code = user_code_parser (user_code)

        IF response = 0, THEN
            DOWHILE READ users_file, "%[^;];%[^;];%[^;];%[^;];%[^;];%[^;];\n", user.session_title,
                user.user_role, user.user_code, user.user_name, user.email, user.password) NOT End Of File
                IF user.user_code = user_code AND user.password = password
                    THEN SET authenticated = 1
                    BREAK
                ENDIF
            ENDDO
        ENDIF

        IF authenticated = 1, THEN
            CLOSE (user_file)
            PRINT("Login successful")
            IF user_role = admin
                THEN void to Admin (user)
            IF user_role = tutor
                THEN void to Tutor (user)
            IF user_role = student
                THEN void to Student (void)
            BREAK
        ELSE
            SET login_attempts = login_attempts + 1
            PRINT ("Incorrect Password or User Code, Please try again. (%d attempts left) ",
                (3 - login_attempts))
        ENDIF
    ENDDO

    IF (login_attempts == max_login_attempts)
        CLOSE (users_file)
        PRINT ("Too many attempts, Try again later.")
    ENDIF
    RETURN authenticated
END

```

```

/* int Admin_menu() */
Admin_menu
    SET choice AS integer

    PRINT ("-----")
    PRINT ("    APU Programming Cafe Management System    ")
    PRINT ("-----")
    PRINT ("                >>  ADMIN PAGE  <<                ")
    PRINT ("-----")
    PRINT ("1. Add User")
    PRINT ("2. View all User")
    PRINT ("3. Add New Programming Session")
    PRINT ("4. Enrol Student in a Session")
    PRINT ("5. Log Out")
    PRINT ("Enter your choice: ")

    GET INPUT choice

    RETURN (choice)
END

/* void Admin (struct users session_user) */
Admin
    SET flag = 1
    SET choice AS integer

    DOWHILE flag = 1
        SET choice = Admin_menu
        CASE OF choice
            1:
                add_User (session_user)
                BREAK

            2:
                View_all_User (session_user)
                BREAK

            3:
                add_New_Session (session_user)
                BREAK

            4:
                Enroll_User_To_Sssion (session_user)
                BREAK

            5:
                SET flag = 0
                PRINT (" ~~~ You are logged out. Thanks for using APU Programming Cafe Management
                System ~~~ ")
                BREAK

            DEFAULT:
                PRINT (" !!! Invalid option, Please try again !!! ")
                PRINT (" -----")

        END CASE
    ENDDO
END

```

```

/* int add_User(struct users session_user) */
add_User (struct users session_user)
    SET structure users AS new_user

    PRINT (" -----")
    PRINT (" |          >> ADMIN PAGE <<          |")
    PRINT (" |          *** ADD USER ***          |")
    PRINT (" -----")

    DOWHILE
        PRINT ("Enter New User's Role (student or tutor): ")
        GET INPUT new_user.user_role
        IF new_user.user_role = Student OR new_user.user_role = Tutor
            BREAK
        ELSE
            PRINT ("Role doesn't exist, Please try again with 'Student' or 'Tutor' only.")
            PRINT ("-----")

        ENDIF
    ENDDO

    DOWHILE
        PRINT ("Enter New User Code [Student (TP) or Tutor (TT)] : ")
        GET INPUT new_user.user_code
        SET INT response = user_code_parser(new_user.user_code)

        IF response = 1, THEN
            BREAK
        ELSE
            PRINT ("Please try again. (Exp: TP123456 or TT123123) ")
            PRINT ("-----")

        ENDIF
    ENDDO

    PRINT ("Enter New User's Name: ")
    GET INPUT new_user.user_name

    PRINT ("Enter New User's Email: ")
    GET INPUT new_user.email

    DOWHILE
        PRINT ("Enter New User's Password (1 Alphabet and 6 character): ")
        GET INPUT new_user.password
        IF new_user.password >= 7, THEN
            BREAK
        ELSE
            PRINT ("Please try again.")
            PRINT ("-----")

        ENDIF
    ENDDO

    OPEN FILE (user_profile.txt) WITH "a" mode
    PRINT (users_file, "NULL;%s;%s;%s;%s;%s", new_user.user_role,
    new_user.user_code, new_user.user_name, new_user.email, new_user.password)
    CLOSE (users_file)

    PRINT ("User Successfully added")
    RETURN 0
END

```

END


```

/* void Print_View_User_Row(char session_title[Max_String], char user_role[Max_String],
    char user_code[Max_User_Code], char user_name[Max_String],
    char email[Max_String], char password[Max_String]) */

Print_View_User_Row(char session_title[Max_String], char user_role[Max_String], char user_code[Max_User_Code], char
user_name[Max_String], char email[Max_String], char password[Max_String]

    PRINT ("| %-20s | %-10s | %-10s | %-10s | %-20s | %-10s |",
        session_title, user_role, user_code, user_name, email, password)

/* int View_all_User (struct users session_user)*/
View_all_User (struct users session_user)

    PRINT(" -----"):
    PRINT(" |          >> ADMIN PAGE <<          |"):
    PRINT(" |      *** VIEW ALL USER ***      |"):
    PRINT(" -----"):

    SET structure users AS u

    SET users_file = OPEN FILE (user_profile.txt) WITH "r" mode

    PRINT("-----|-----|-----|-----|-----|")
    PRINT("| %-20s | %-10s | %-10s | %-10s | %-20s | %-10s | \n", "Title", "User Role", "User Code", "User Name",
    "Email", "Password")
    PRINT("-----|-----|-----|-----|-----|")

    DOWHILE (fscanf(users_file, "%[^;];%[^;];%[^;];%[^;];%[^;];%[^;];\n", u.session_title, u.user_role,
        u.user_code, u.user_name, u.email, u.password) NOT End Of File
        Print_View_User_Row (u.session_title, u.user_role, u.user_code, u.user_name, u.email, u.password)
    ENDDO
    CLOSE FILE
    RETURN 0
END

```

```

/* int add_New_Session(struct users session_user)*/
add_New_Session (struct users session_user)

    PRINT (" -----");
    PRINT (" |          >> ADMIN PAGE <<          |");
    PRINT (" |      *** ADD NEW SESSION ***      |");
    PRINT (" -----");

    SET session_file = OPEN FILE (default_session.txt) WITH "a+" mode
    SET enrol_session_file = OPEN FILE (enrol_session.txt) WITH "r" mode
    SET users_file = OPEN FILE (user_profile.txt) WITH "r" mode

    SET struct sessions AS session
    SET add_session WITH MAXIMUM LENGTH Max_Session_Code
    SET user_code WITH MAXIMUM LENGTH Max_Session_Code
    SET session_code WITH MAXIMUM LENGTH Max_Session_Code

    DOWHILE
        PRINT ("Enter New Session's Code (6 characters): ")
        GET INPUT session_code
        IF STRING LENGTH ( session_code) = 6
            BREAK
        ELSE
            PRINT ("Please try again")
        ENDIF

        DOWHILE (fscanf(session_file, "%[^,];%[^,];%[^,];%[^,];%[^,];%[^,];\n", session.session_code,
            session.title, session.day, session.start_time, session.location, session.tutor_code) NOT End Of File
            IF session.session_code , session_code = 0
                PRINT ("The Session has already exists")
                RETURN 0
            ENDIF
        ENDDO

        PRINT ("Enter New Session's Name: ")
        GET INPUT session.title

        PRINT ("Enter New Session's Day: ")
        GET INPUT session.day

        PRINT ("Enter New Session's start time: ")
        GET INPUT session.start_time

        PRINT ("Enter New Session's Location: ")
        GET INPUT session.location

        PRINT ("Tutor who handle this session (Please enter 6 digit Tutor Code start with 'TT'): ")
        GET INPUT user_code
        SET INT response = user_code_parser(user_code)

```

```

IF response = 0
    SET structure users AS user
    SET flag = 0
    DOWHILE (fscanf(users_file, "%[^:];%[^:];%[^:];%[^:];%[^:];%[^:];\n", user.session_title,
user.user_role, user.user_code, user.user_name, user.email, user.password) NOT End Of File
        IF user.user_code , user_code = 0
            SET flag = 1
            BREAK
        ENDIF
    ENDDO
    IF flag NOT = 1
        PRINT ("Tutor does not exists")
        CLOSE FILE users_file
        RETURN 0
    ENDIF

    SET struct enrol_sessions AS es
    SET flag2 = 0
    DOWHILE (fscanf(enrol_session_file, "%[^:];%[^:];%[^:];%[^:];\n",
es.user_code, es.user_name, es.user_role, es.session_code) NOT End Of File
        IF es.user_code = user_code AND es.session_code = session_code
            SET flag2 = 1
            BREAK
        ENDIF
    ENDDO
    IF flag2 = 0
        PRINT ("Tutor has already enrolled in the session")
        CLOSE FILE enrol_session_file
        RETURN 0
    ENDIF
    ENDDO
    PRINT (session_file, "%s;%s;%s;%s;", session.title, session.day, session.start_time, session.location)
    CLOSE FILE (session_file)
    RETURN 0
END
END

```

```

/* int Enroll_User_to_Session (struct users session_user) */
Enroll_User_to_Session (struct users session_user)

    PRINT ("-----")
    PRINT (" |          >> ADMIN PAGE <<          |")
    PRINT (" |      *** ENROLL USER TO SESSION ***      |")
    PRINT ("-----")

    SET default_session_file = OPEN FILE default_session.txt WITH "a+" mode
    SET users_file = OPEN FILE user_profile.txt WITH "a+" mode

    PRINT ("Enter a session code to enrol user to that session: ")
    SET session_code WITH MAXIMUM LENGTH Max_Session_Code
    GET INPUT session_code

    SET structure sessions AS s
    SET FoundSession = 0

    DOWHILE (fscanf(default_session_file, "%[^:];%[^:];%[^:];%[^:];%[^:];%[^:];", s.session_code,
        s.title, s.day, s.start_time, s.location, s.tutor_code) NOT End Of File
        IF s.session_code , s.session_code = 0
            SET FoundSession = 1
            BREAK
        ENDIF
    ENDDO

    IF FoundSession == 0
        PRINT ("This session does not exist. ")
        RETURN 0
    ENDIF

    PRINT ("Enter the Enter the student code to enrol him/her in this session: ")
    SET user_code WITH MAXIMUM LENGTH Max_User_Code
    GET INPUT user_code

    SET structure users AS u
    SET integer FoundUser = 0

    DOWHILE (fscanf(users_file, "%[^:];%[^:];%[^:];%[^:];%[^:];%[^:];\n", u.session_title,
        u.user_role, u.user_code, u.user_name, u.email, u.password) NOT End Of File
        IF u.user_code , user_code = 0
            SET FoundUser = 1
            BREAK
        ENDIF
    ENDDO

    IF FoundUser = 0
        PRINT ("This user does not exist.")
        RETURN 0
    ENDIF

    IF u.user_role = tutor
        PRINT ("Tutor cannot be enrolled here.")
    ENDIF

    SET enrol_session_file = OPEN FILE (enrol_session.txt) WITH "a+" mode

```

```

SET structure enrol_session AS es
DOWHILE (fscanf(enrol_session_file, "%[^,];%[^,];%[^,];%[^,];",
               es.user_code, es.user_name, es.user_role, es.session_code) NOT End Of File
    IF es.user_code = user_code AND es.session_code = session_code
        PRINT ("This user already enrolled")
        RETURN 0
    ENDIF
ENDDO
PRINT (enrol_session_file, "%s;%s;%s;%s; u.user_code, u.user_name, u.user_role, s.session_code)
PRINT ("Student successfully added in this session ~")
CLOSE FILE (default_session_file)
CLOSE FILE (users_file)
CLOSE FILE (enrol_session_file)
RETURN 0
END

/* int Tutor_menu() */
Tutor_menu

    SET choice AS INTEGER
    PRINT ("-----")
    PRINT (" |   APU Programming Cafe Management System   |")
    PRINT ("-----")
    PRINT (" |               >>  TUTOR PAGE  <<               |")
    PRINT ("-----")
    PRINT ("1. View my sessions")
    PRINT ("2. View students in sessions")
    PRINT ("3. Log Out")
    PRINT ("Enter your choice")

    GET INPUT choice

    RETURN (choice)
END

```

```

/* void Tutor (struct users session_user)*/
Tutor(struct users session_user)

    SET flag = 1
    SET choice AS INTEGER

    DOWHILE flag = 1
        SET choice = Tutor_menu()
        CASE OF choice
            1:
                View_Tutor_Session (session_user)
                BREAK
            2:
                View_Students_in_Session (session_user)
                BREAK
            3:
                SET flag = 0
                PRINT (" ~ You are logged out. Thanks for using APU Programming Cafe
                Management System ~ ")
            DEFAULT:
                PRINT (" !!! Invalid option, Please try again !!! )
        END CASE
    ENDDO
END

/* int View_Tutor_Session (struct users session_user) */
View_Tutor_Session (struct users session_user)

    PRINT (" -----")
    PRINT (" |          >> TUTOR PAGE <<          |")
    PRINT (" |          *** MY SESSION ***          |")
    PRINT (" -----")

    SET struct enrol_sessions AS s

    SET enrol_session_file = OPEN FILE (enrol_session.txt) WITH "r" mode
    DOWHILE (fscanf(enrol_session_file, "%[^;];%[^;];%[^;];%[^;];", s.user_code,
        s.user_name, s.user_role, s.session_code) NOT EOF

        IF s.user_code = session_user.user_code
            PRINT ("This is your sessions.")
            PRINT ("%s %s %s %s ", s.user_code, s.user_name, s.user_role, s.session_code)
            BREAK
        ENDIF
    ENDDO
    CLOSE FILE
    RETURN 0
END

```

```

J
/* int View_Students_in_Session(struct users session_user)*/
View_Students_in_Session (struct users session_user)

    PRINT ("-----")
    PRINT ("|                >>  TUTOR PAGE  <<                |")
    PRINT ("|    ***  STUDENT ENROLLED IN MY SESSION  ***    |")
    PRINT ("-----")

    SET enrol_session_file = OPEN FILE enrol_session.txt WITH "r" mode
    SET default_session_file = OPEN FILE default_session.txt WITH "r" mode

    SET struct enrol session AS es
    SET struct sessions AS s

    SET foundSession = 0
    DOWHILE (fscanf(default_session_file, "%[^:];%[^:];%[^:];%[^:];%[^:];%[^:];\n", s.session_code, s.title, s.day,
        s.start_time, s.location, s.tutor_code) != EOF)
        IF session_user.user_code = s.tutor_code
            BREAK
        ENDIF
    ENDDO
    IF FoundSession = 0
        PRINT ("You have not enrolled in any session. ")
        RETURN 0
    ENDIF

    DOWHILE (fscanf(enrol_session_file, "%[^:];%[^:];%[^:];%[^:];\n", es.user_code,
        es.user_name, es.user_role, es.session_code) NOT EOF)
        IF es.user_role = student AND es.session_code = s.session_code
            PRINT ("%s %s %s\n", es.user_code, es.user_name, es.session_code)

            REWIND THE FILE POINTER TO BY default_session_file TO THE BIGINING

            DOWHILE (fscanf(default_session_file, "%[^:];%[^:];%[^:];%[^:];%[^:];%[^:];\n",
                s.session_code, s.title, s.day, s.start_time, s.location, s.tutor_code) NOT EOF)
                IF s.session_code = " "
                    PRINT ("%s ", s.location)
                    BREAK
                ENDIF
            ENDDO
        ENDIF
    ENDDO

    CLOSE FILE enrol_session_file
    CLOSE FILE default_session_file
    RETURN 0
END

```

```

/* int Student_menu()*/
Studenty_munu
    SET choice AS integer

    PRINT ("-----")
    PRINT (" |   APU Programming Cafe Management System   |")
    PRINT (" |-----|")
    PRINT (" |           >>  STUDENT PAGE  <<           |")
    PRINT (" |-----|")
    PRINT ("1. View my session. ")
    PRINT ("2. Enrol into session. ")
    PRINT ("3. Log Out. ")
    PRINT ("Enter your choice: ")

    GET INPUT choice

    RETURN choice
END

/* void Student() */
Student(struct users session_user)
    SET flag = 1
    SET choice AS INTEGER

    DOWHILE flag = 1
        SET choice = Student_menu
        CASE of choice
            1.
                View_Student_Session(session_user)
                BREAK
            2.
                Enroll_into_Session(session_user)
                BREAK
            3.
                SET flag = 0
                PRINT ("You are logged out. Thanks for using APU Programming Cafe
                Management System ~~~")
                BREAK
        DEFAULT:
            PRINT ("!!! Invalid option, Please try again !!! ")
        END CASE
    ENDDO
END

```



```

/* int View_Student_Session(struct users session_user) */
View_Student_Session(struct users session_user)

    PRINT ("-----")
    PRINT (" |          >> STUDENT PAGE <<          |")
    PRINT (" |          *** VIEW MY SESSION ***          |")
    PRINT ("-----")

    SET structure enrol session AS es

    SET enrol_session_file = OPEN FILE enrol_session.txt WITH "r" mode

    DOWHILE (fscanf(enrol_session_file, "%[^;];%[^;];%[^;];%[^;]\n", es.user_code,
        es.user_name, es.user_role, es.session_code) NOT EOF)

        IF es.user_code, session_user.user_code = 0
            PRINT ("This is your sessions.")
            PRINT ("%s %s %s %s\n", es.user_code, es.user_name, es.user_role,
                es.session_code)
            BREAK
        ENDIF
    ENDDO

    CLOSE FILE enrol_session_file
    RETURN 0
END

```

```

/* int Enroll_into_Session(struct users session_user)*/
Enroll_into_Session(struct users session_user)

    PRINT ("-----")
    PRINT ("|          >> STUDENT PAGE <<          |")
    PRINT ("|          *** MY ENROLLED SESSION ***    |")
    PRINT ("-----")

    SET struct enrol_session AS new_es

    SET enrol_session_file = OPEN FILE enrol_session.txt WITH "a+" mode

    SET session_code WITH A MAXIMUM LENGTH Max_Session_Code

    PRINT ("Which session you want to enrol? ")
    PRINT ("Please enter the session code: ")
    GET INPUT session_code

    DOWHILE (fscanf(enrol_session_file, "%[^;];%[^;];%[^;];%[^;];\n", new_es.user_code,
        new_es.user_name, new_es.user_role, new_es.session_code) NOT EOF
        IF new_es.session_code = session_code AND new_es.user_code = session_user.user_code
            PRINT ("You are already in this session")
            RETURN 0
        ENDIF
    ENDDO

    PRINT (enrol_session_file, "%s;%s;%s;%s;", session_user.user_code, session_user.user_name,
        session_user.user_role, session_code)
    CLOSE FILE enrol_session_file
    PRINT ("You have enrolled into a new session!!! ")
    PRINT ("This is you enrolled sessions.")
    PRINT ("%s %s %s %s ", session_user.user_code, session_user.user_name, session_user.user_role,
        session_code)
    RETURN 0
END

```

```

/* void PrintTableRow(char session_code[Max_Session_Code], char title[Max_String], char day[Max_String],
char start_time[Max_String], char location[Max_String], char tutor_code[Max_User_Code]*/

PrintTableRow(char session_code[Max_Session_Code], char title[Max_String], char day[Max_String],
char start_time[Max_String], char location[Max_String], char tutor_code[Max_User_Code])

    PRINT ("| %-12s | %-25s | %-10s | %-10s | %-10s | %-11s |\n", session_code, title, day, start_time,
location, tutor_code)

/* int Session_menu()*/
Session_munu

    PRINT ("-----")
    PRINT ("|   APU Programming Cafe Management System   |")
    PRINT ("|-----|")
    PRINT ("|               >>  Available Sessions  <<               |")
    PRINT ("|-----")

    SET structure sessions AS s
    OPEN FILE default_session.txt WITH "r" mode
    DOWHILE (fscanf(default_session_file, "%[^;];%[^;];%[^;];%[^;];%[^;];%[^;];\n", s.session_code, s.title, s.day,
s.start_time, s.location, s.tutor_code) != EOF)
        PRINT ("%s %s %s %s %s %s\n", s.session_code, s.title, s.day, s.start_time, s.location, s.tutor_code)
    ENDDO

    FILE CLOSE default_session_file
    RETURN 0

END

/* int create_file()*/
create_file

    SET default_session_file = OPEN FILE default_session.txt WITH "w" mode
    SET enrol_session_file = OPEN FILE enrol_session.txt WITH "w" mode
    SET users_file = OPEN FILE user_profile.txt WITH "w" mode

    SET default_session =
        [
        ["PYP101", "Python Programming", "Saturday", "9.00am", "C-01-01", "TU265663"]
        ["JAV102", "Java Programming", "Sunday", "9.00am", "C-01-02", "TU009650"]
        ["CSC103", "C Programming", "Saturday", "2.00pm", "C-01-03", "TU544654"]
        ["WEB104", "Web Development", "Sunday", "2.00pm", "C-01-04", "TU577001"]
        ["CSP105", "C Sharp Programming", "Monday", "7.00pm", "C-01-05", "TU683357"]
        ]

    DO x = 0 TO 5
        DO y = 0 TO 6
            FILE STORE "{default session[x][y]};" TO default_sessions_file
        ENDDO
        FILE STORE "\n" TO default_sessions_file
    ENDDO

```

```

SET enrol_session =
    [
        ["TT123456", "Danny", "tutor", "PYP101"],
        ["TT789123", "YitWah", "tutor", "JAV102"],
        ["TT456789", "Joanne", "tutor", "CPL103"},
        ["TT101112", "Linda", "tutor", "WEB104"],
        ["TT131415", "William", "tutor", "CSP105"],
        ["TP987654", "Jia Xin", "student", "CPL103"],
        ["TP321987", "Man Wei", "student", "PYP101"],
        ["TP654321", "Wong Xuen", "student", "CSP105"]
    ]

DO x = 0 TO 8
    DO y = 0 TO 4
        FILE STORE "{enrol_session[x][y]};" TO enrol_sessions_file
    ENDDO
    FILE STORE "\n" TO enrol_sessions_file
ENDDO

SET users =
    [
        ["NULL", "admin", "AD111222", "Susan", "susan1@apu.edu.my", "Y111222"],
        ["NULL", "admin", "AD333444", "Anna", "anna2@apu.edu.my", "Y333444"],
        ["NULL", "admin", "AD555666", "Bryan", "bryan3@apu.edu.my", "Y555666"],

        ["PYTHON PROGRAMMING", "tutor", "TT123456", "Danny", "danny1@apu.edu.my", "X123456"],
        ["JAVA PROGRAMMING", "tutor", "TT789123", "YitWah", "yitwah2@apu.edu.my", "X789123"],
        ["C Programming", "tutor", "TT456789", "Joanne", "joanne3@apu.edu.my", "X456789"],
        ["Web Development", "tutor", "TT101112", "Linda", "linda4@apu.edu.my", "X101112"],
        ["C Sharp Programming", "tutor", "TT131415", "William", "william5@apu.edu.my", "X131415"],

        ["NULL", "student", "TP987654", "Jiaxin", "jiaxin1@apu.edu.my", "Z987654"],
        ["NULL", "student", "TP321987", "Manwei", "manwei2@apu.edu.my", "Z321987"],
        ["NULL", "student", "TP654321", "Wongxuen", "wongxuen@apu.edu.my", "Z654321"]
    ]

DO x = 0 TO 11
    DO y = 0 TO 6
        FILE STORE "{users[x][y]};" TO users_file
    ENDDO

DO x = 0 TO 11
    DO y = 0 TO 6
        FILE STORE "{users[x][y]};" TO users_file
    ENDDO

    FILE STORE "\n" TO users_file
ENDDO

CLOSE FILE default_session_file
CLOSE FILE enrol_session_file
CLOSE FILE users_file
RETURN 0
END

```

```

/* int user_code_parser(char *user_code) */
User_code_parser(char *user_code)

    SET valid_user_id = 0
    IF ((user_code[0] == 'A' AND user_code[1] == 'D') ||
        (user_code[0] == 'a' AND user_code[1] == 'd') ||
        (user_code[0] == 'A' AND user_code[1] == 'D') ||
        (user_code[0] == 'a' AND user_code[1] == 'd'))

        user_code[0] = 'A';
        user_code[1] = 'D';

        SET valid_user_id = 1

    ELSE IF ((user_code[0] == 'T' AND user_code[1] == 'T') ||
        (user_code[0] == 't' AND user_code[1] == 't') ||
        (user_code[0] == 't' AND user_code[1] == 'T') ||
        (user_code[0] == 'T' AND user_code[1] == 't'))

        user_code[0] = 'T';
        user_code[1] = 'T';

        SET valid_user_id = 1

    ELSE IF ((user_code[0] == 'T' AND user_code[1] == 'P') ||
        (user_code[0] == 't' AND user_code[1] == 'p') ||
        (user_code[0] == 't' AND user_code[1] == 'P') ||
        (user_code[0] == 'T' AND user_code[1] == 'p'))

        user_code[0] = 'T';
        user_code[1] = 'P';

        SET valid_user_id = 1

    ENDIF
    RETURN valid_user_id
END

```

3. Sample output

3.1 Main Menu

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  HOME PAGE  <<              |  
|-----|  
  
1. Login  
2. View Available Programming Session  
3. Exit the Programming Cafe Session  
Enter your choice: 
```

Main Menu

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  HOME PAGE  <<              |  
|-----|  
  
1. Login  
2. View Available Programming Session  
3. Exit the Programming Cafe Session  
Enter your choice: 10  
!!! Invalid option, Please try again !!!  
  
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  HOME PAGE  <<              |  
|-----|  
  
1. Login  
2. View Available Programming Session  
3. Exit the Programming Cafe Session  
Enter your choice:
```



Main Menu – Check if the user input an available option, if no system will ask the user to try again

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> HOME PAGE <<              |  
|-----|  
  
1. Login  
2. View Available Programming Session  
3. Exit the Programming Cafe Session  
Enter your choice:  
  
█
```

Main Menu – If the user did not enter any option but just press on ‘Enter’

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> HOME PAGE <<              |  
|-----|  
  
1. Login  
2. View Available Programming Session  
3. Exit the Programming Cafe Session  
Enter your choice: 1  
  
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> LOGIN PAGE <<              |  
|-----|  
  
Please Enter Your User Code: █
```

Main Menu (Entered valid option) - Turn into Login Page

3.2 Login Page

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> LOGIN PAGE <<              |  
|-----|  
Please Enter Your User Code: █
```

Login Page

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> LOGIN PAGE <<              |  
|-----|  
Please Enter Your User Code: Yap JX  
Password:  
Incorrect Password or User Code, Please try again. (2 attempts left)  
Please Enter Your User Code: █
```

Login Page – Check if the user enters a valid User Code

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> LOGIN PAGE <<              |  
|-----|  
Please Enter Your User Code: Yap JX  
Password: Incorrect Password or User Code, Please try again. (2 attempts left)  
  
Please Enter Your User Code: Tp123456  
Password: 123145  
Incorrect Password or User Code, Please try again. (1 attempts left)  
Please Enter Your User Code: █
```

Login Page – Login Attempts will decrease if the user enters wrong User Code or Password


```

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> LOGIN PAGE <<              |
|-----|

Please Enter Your User Code: Yap JX
Password: Incorrect Password or User Code, Please try again. (2 attempts left)

Please Enter Your User Code: Tp01235
Password: 12566
Incorrect Password or User Code, Please try again. (1 attempts left)

Please Enter Your User Code: blablaba
Password: idk
Incorrect Password or User Code, Please try again. (0 attempts left)

Too many attempts, Try again later.

Process finished with exit code 0

```

Login Page – User has only 3 attempts to login, if there is no more attempt to login, the program will end directly.

```

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> LOGIN PAGE <<              |
|-----|

Please Enter Your User Code: AD111222
Password: Y111222
Login successful

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> ADMIN PAGE <<              |
|-----|

1. Add User
2. View all User
3. Add New Programming Session
4. Enroll User to a Session
5. Log Out
Enter your choice: █

```

Login to Admin Page

```

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> LOGIN PAGE <<              |
|-----|

Please Enter Your User Code: TT123456
Password: X123456
Login successful

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> TUTOR PAGE <<              |
|-----|

1. View my sessions
2. View students in sessions
3. Log Out
Enter your choice: █

```

Login to Tutor Page

```

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> LOGIN PAGE <<              |
|-----|

Please Enter Your User Code: TP987654
Password: Z987654
Login successful

-----
|      APU Programming Cafe Management System      |
|-----|
|              >> STUDENT PAGE <<              |
|-----|

1. View my sessions
2. Enroll into session
3. Log Out
Enter your choice: █

```

Login to Student Page

Login page (Entered correct User Code and Password) – Login Successfully and turn into respective page

3.3 Session Menu

```

-----
|      APU Programming Cafe Management System      |
|-----|
|          >> Available Sessions <<          |
|-----|

```

Session Code	Title	Day	Time	Location	Tutor ID
PYP101	PYTHON PROGRAMMING	Saturday	9.00am	C-01-01	TT123456
JAV102	JAVA PROGRAMMING	Sunday	9.00am	C-01-02	TT789123
CPL103	C PROGRAMMING	Saturday	2.00pm	C-01-03	TT456789
WEB104	WEB DEVELOPMENT	Sunday	2.00pm	C-01-04	TT101112
CSP105	C SHARP PROGRAMMING	Monday	7.00pm	C-01-05	TT131415

Session Menu – Show

3.4 Admin Menu

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  ADMIN PAGE  <<              |  
-----  
1. Add User  
2. View all User  
3. Add New Programming Session  
4. Enroll User to a Session  
5. Log Out  
Enter your choice: █
```

Admin Page

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  ADMIN PAGE  <<              |  
-----  
1. Add User  
2. View all User  
3. Add New Programming Session  
4. Enroll User to a Session  
5. Log Out  
Enter your choice: 10  
!!! Invalid option, Please try again !!!  
  
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  ADMIN PAGE  <<              |  
-----  
1. Add User  
2. View all User  
3. Add New Programming Session  
4. Enroll User to a Session  
5. Log Out  
Enter your choice: █
```



Admin Page – Check if the user input a valid option, if no system will ask the user to try again

4.4.1 Add User

```
-----  
|                >> ADMIN PAGE <<                |  
|                *** ADD USER ***                |  
-----  
  
Enter New User's Role (student or tutor): student  
Enter New User Code [Student (TP) or Tutor (TT)] : Tp066475  
Enter New User's Name: YJX  
Enter New User's Email: YJX0211@gmail.com  
Enter New User's Password (1 Alphabet and 6 character): Y030211  
~~~ User successfully added ~~~
```

Admin Page (Add user) – Add a new user by entering their role, user code, name, email, and password

```
NULL;admin;AD111222;Susan;susan1@apu.edu.my;Y111222;  
NULL;admin;AD333444;Anna;anna2@apu.edu.my;Y333444;  
NULL;admin;AD555666;Bryan;bryan3@apu.edu.my;Y555666;  
PYTHON PROGRAMMING;tutor;TT123456;Danny;danny1@apu.edu.my;X123456;  
JAVA PROGRAMMING;tutor;TT789123;YitWah;yitwah2@apu.edu.my;X789123;  
C Programming;tutor;TT456789;Joanne;joanne3@apu.edu.my;X456789;  
Web Development;tutor;TT101112;Linda;linda4@apu.edu.my;X101112;  
C Sharp Programming;tutor;TT131415;William;william5@apu.edu.my;X131415;  
NULL;tutor;TT121212;Jennie;Jennie6@apu.edu.my;X121212;  
NULL;student;TP987654;Jia xin;jiaxin1@apu.edu.my;Z987654;  
NULL;student;TP321987;Man wei;manwei2@apu.edu.my;Z321987;  
NULL;student;TP654321;Wong xuen;wongxuen@apu.edu.my;Z654321;  
NULL;student;TP066475;YJX;YJX0211@gmail.com;Y030211;|
```



The user (student) is added successfully into user profile file

4.4.2 View All User

```

-----
|                >> ADMIN PAGE <<                |
|                *** VIEW ALL USER ***                |
|-----|-----|-----|-----|-----|-----|
| Title          | User Role | User Code | User Name | Email          | Password |
|-----|-----|-----|-----|-----|-----|
| NULL           | admin     | AD111222 | Susan    | susan1@apu.edu.my | Y111222 |
| NULL           | admin     | AD333444 | Anna     | anna2@apu.edu.my | Y333444 |
| NULL           | admin     | AD555666 | Bryan    | bryan3@apu.edu.my | Y555666 |
| PYTHON PROGRAMMING | tutor     | TT123456 | Danny    | danny1@apu.edu.my | X123456 |
| JAVA PROGRAMMING  | tutor     | TT789123 | YitWah   | yitwah2@apu.edu.my | X789123 |
| C Programming     | tutor     | TT456789 | Joanne    | joanne3@apu.edu.my | X456789 |
| Web Development   | tutor     | TT101112 | Linda     | linda4@apu.edu.my | X101112 |
| C Sharp Programming | tutor     | TT131415 | William   | william5@apu.edu.my | X131415 |
| NULL             | tutor     | TT121212 | Jennie    | Jennie6@apu.edu.my | X121212 |
| NULL             | student   | TP987654 | Jia xin   | jiaxin1@apu.edu.my | Z987654 |
| NULL             | student   | TP321987 | Man wei   | manwei2@apu.edu.my | Z321987 |
| NULL             | student   | TP654321 | Wong xuen | wongxuen@apu.edu.my | Z654321 |

```

Admin Page (View All User)

```

-----
|                >> ADMIN PAGE <<                |
|                *** VIEW ALL USER ***                |
|-----|-----|-----|-----|-----|-----|
| Title          | User Role | User Code | User Name | Email          | Password |
|-----|-----|-----|-----|-----|-----|
| NULL           | admin     | AD111222 | Susan    | susan1@apu.edu.my | Y111222 |
| NULL           | admin     | AD333444 | Anna     | anna2@apu.edu.my | Y333444 |
| NULL           | admin     | AD555666 | Bryan    | bryan3@apu.edu.my | Y555666 |
| PYTHON PROGRAMMING | tutor     | TT123456 | Danny    | danny1@apu.edu.my | X123456 |
| JAVA PROGRAMMING  | tutor     | TT789123 | YitWah   | yitwah2@apu.edu.my | X789123 |
| C Programming     | tutor     | TT456789 | Joanne    | joanne3@apu.edu.my | X456789 |
| Web Development   | tutor     | TT101112 | Linda     | linda4@apu.edu.my | X101112 |
| C Sharp Programming | tutor     | TT131415 | William   | william5@apu.edu.my | X131415 |
| NULL             | tutor     | TT121212 | Jennie    | Jennie6@apu.edu.my | X121212 |
| NULL             | student   | TP987654 | Jia xin   | jiaxin1@apu.edu.my | Z987654 |
| NULL             | student   | TP321987 | Man wei   | manwei2@apu.edu.my | Z321987 |
| NULL             | student   | TP654321 | Wong xuen | wongxuen@apu.edu.my | Z654321 |
| NULL             | student   | TP066475 | YJX       | YJX030211@gmail.com | Y030211 |

```

Admin Page (View All User) – Student YJX who added by admin in Admin Add User page are show below of the table.

4.4.3 Add New Session

```
-----  
|           >> ADMIN PAGE <<           |  
|           *** ADD NEW SESSION ***       |  
-----  
Enter New Session's Code (6 characters): IN6666  
Enter New Session's Name: IntroductionToNetworking  
Enter New Session's Day: Testing  
Enter New Session's start time: Testing  
Enter New Session's Location: Testing  
Tutor who handle this session (Please enter 6 digit Tutor Code start with 'TT'): TT066475  
Tutor does not exists.
```

Admin Page (Add New Session) –
Session added unsuccessfully because this tutor does not exists.

```
-----  
|           >> ADMIN PAGE <<           |  
|           *** ADD NEW SESSION ***       |  
-----  
Enter New Session's Code (6 characters): IN6666  
Enter New Session's Name: IntroductionToNetworking  
Enter New Session's Day: Testing  
Enter New Session's start time: Testing  
Enter New Session's Location: Testing  
Tutor who handle this session (Please enter 6 digit Tutor Code start with 'TT'): TT101112  
This tutor has already enrolled in another session.
```

Admin Page (Add New Session) –
Session added unsuccessfully because Tutor who name Linda with tutor code TT101112 are
already enrolled in another session which is Web Development session.

4.4.4 Enrol Student to Session

```

-----
|                >>  ADMIN PAGE  <<                |
|                ***  ENROLL STUDENT TO SESSION  ***  |
|-----|
Enter a session code to enroll student to that session: PYP101
Enter the student code to enroll him/her in this session: TP987654
Student successfully added in this session ~~

```

Admin Page (Enrol Student to Session) – Student enrol successfully into PYP101 session

```

TT123456;Danny;tutor;PYP101;
TT789123;YitWah;tutor;JAV102;
TT456789;Joanne;tutor;CPL103;
TT101112;Linda;tutor;WEB104;
TT131415;William;tutor;CSP105;
TP987654;Jia Xin;student;CPL103;
TP321987;Man Wei;student;PYP101;
TP654321;Wong Xuen;student;CSP105;
TP987654;Jia xin;student;PYP101;|

```

The student who names Jia Xin is successfully enrol into session PYP101 by admin and it will show below of the enrol session file

```

-----
|                >>  ADMIN PAGE  <<                |
|                ***  ENROLL STUDENT TO SESSION  ***  |
|-----|
Enter a session code to enroll student to that session: WEB104
Enter the student code to enroll him/her in this session: TP987654
Student successfully added in this session ~~

-----
|                >>  ADMIN PAGE  <<                |
|                ***  ENROLL STUDENT TO SESSION  ***  |
|-----|
Enter a session code to enroll student to that session: CSP105
Enter the student code to enroll him/her in this session: TP987654
Student successfully added in this session ~~

```

```

-----
|                >>  ADMIN PAGE  <<                |
|                ***  ENROLL STUDENT TO SESSION  ***  |
|-----|
Enter a session code to enroll student to that session: JAV102
Enter the student code to enroll him/her in this session: TP987654
Student successfully added in this session

-----
|                >>  ADMIN PAGE  <<                |
|                ***  ENROLL STUDENT TO SESSION  ***  |
|-----|
Enter a session code to enroll student to that session: JAV102
Enter the student code to enroll him/her in this session: TP987654
This user already enrolled.

```

Student Jia Xin can also be enrolled into other any session unless Jia Xin have already enrolled in that session.

```
TT123456;Danny;tutor;PYP101;  
TT789123;YitWah;tutor;JAV102;  
TT456789;Joanne;tutor;CPL103;  
TT101112;Linda;tutor;WEB104;  
TT131415;William;tutor;CSP105;  
TP987654;Jia Xin;student;CPL103;  
TP321987;Man Wei;student;PYP101;  
TP654321;Wong Xuen;student;CSP105;  
TP987654;Jia xin;student;PYP101;  
TP987654;Jia xin;student;WEB104;  
TP987654;Jia xin;student;CSP105;  
TP987654;Jia xin;student;JAV102;
```

Admin has enrol student Jia Xin into 4 additional session which is PYP101, WEB104, CSP105 and JAV102

3.5 Tutor Menu

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  TUTOR PAGE  <<              |  
-----  
1. View my sessions  
2. View students in sessions  
3. Log Out  
Enter your choice:
```

Tutor Page

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  TUTOR PAGE  <<              |  
-----  
1. View my sessions  
2. View students in sessions  
3. Log Out  
Enter your choice: 5  
!!! Invalid option, Please try again !!!  
  
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  TUTOR PAGE  <<              |  
-----  
1. View my sessions  
2. View students in sessions  
3. Log Out  
Enter your choice:
```



Tutor Page – Check if the user input a valid option, if no system will ask the user to try again

4.5.1 View Tutor Session

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  TUTOR PAGE  <<              |  
-----  
1. View my sessions  
2. View students in sessions  
3. Log Out  
Enter your choice: 1  
  
-----  
|              >>  TUTOR PAGE  <<              |  
|              ***  MY SESSION  ***              |  
-----  
This is your sessions.  
TT123456 Danny tutor PYP101
```

Tutor Page (View My Session) – Tutor can view their teaching session by choosing option 1

4.5.2 View Student in Session

```
-----  
|      APU Programming Cafe Management System      |  
-----  
|              >>  TUTOR PAGE  <<              |  
-----  
1. View my sessions  
2. View students in sessions  
3. Log Out  
Enter your choice: 2  
  
-----  
|              >>  TUTOR PAGE  <<              |  
|              ***  STUDENT ENROLLED IN MY SESSION  ***              |  
-----  
TP321987 Man Wei PYP101 C-01-01
```

Tutor Page (View My Session) – Tutor can view which student are joining his/her session by choosing option 2

3.6 Student Menu

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  STUDENT PAGE  <<              |  
|-----|  
  
1. View my sessions  
2. Enroll into session  
3. Log Out  
Enter your choice: 
```

Student Menu

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  STUDENT PAGE  <<              |  
|-----|  
  
1. View my sessions  
2. Enroll into session  
3. Log Out  
Enter your choice: 4  
!!! Invalid option, Please try again !!!  
  
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >>  STUDENT PAGE  <<              |  
|-----|  
  
1. View my sessions  
2. Enroll into session  
3. Log Out  
Enter your choice: 
```



Student Page – Check if the user input a valid option, if no system will ask the user to try again

4.6.1 View Student Session

```
-----  
|      APU Programming Cafe Management System      |  
|-----|  
|              >> STUDENT PAGE <<              |  
|-----|  
1. View my sessions  
2. Enroll into session  
3. Log Out  
Enter your choice: 1  
-----  
|              >> STUDENT PAGE <<              |  
|              *** VIEW MY SESSION ***           |  
|-----|  
This is your sessions.This is your sessions.  
TP654321 Wong Xuen student CSP105
```

Student Page (View student session) – Student can view their session that enrolled by choosing option 1

4.6.2 Enrol into Session

```
-----  
|                >> STUDENT PAGE <<                |  
|                *** MY ENROLLED SESSION ***          |  
-----  
Which session you want to enroll?  
Please enter the session code: PYP101  
  
You have enrolled in to a new session!!!  
  
This is you enrolled sessions.  
TP654321 Wong xuen student PYP101
```

Student Page (Enrol into session) – Beside enrolled by Admin, Student can also enrol themselves to any available session

```
-----  
|                >> STUDENT PAGE <<                |  
|                *** VIEW MY SESSION ***            |  
-----  
This is your sessions.This is your sessions.  
TP654321 Wong Xuen student CSP105  
TP654321 Wong xuen student PYP101
```

Student page (Enrol into Session) – if student success to enrol themselves into available session, they can see their “view session” added a new session that they have enrolled.

4. Conclusion

In conclusion, the goal of this “Programming Café “ that established by APU is to enhance APU students’ problem-solving skills, boost their confidence in coding and improve their performance in coding. In order to meet these goals, this “Programming Café” have provided extra coding sessions for student to join and improve themselves in programming language. Every session are handle by selective APU Graduate Students who having good knowledge in their perspective programming language so that gain the trust from the students.